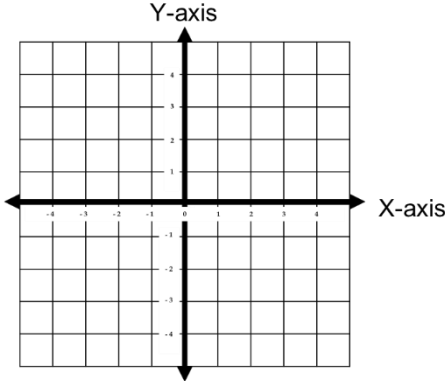
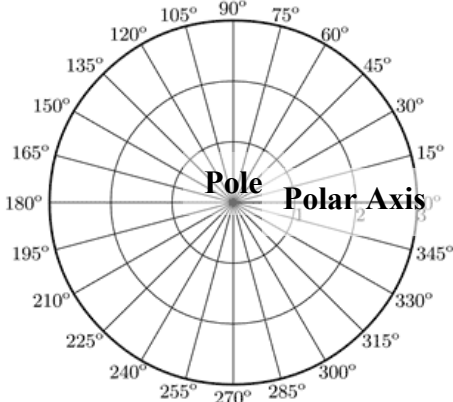


5.2 Polar Coordinates

<i>Cartesian (Rectangular) Coordinates</i>	<i>Polar Coordinates</i>
- Plot points by moving - horizontally (x-direction) and - vertically (y-direction)  - We have coordinates (x, y)	- Plot points by - rotating by a given angle and - moving a specified length  - We have coordinates (r, θ)

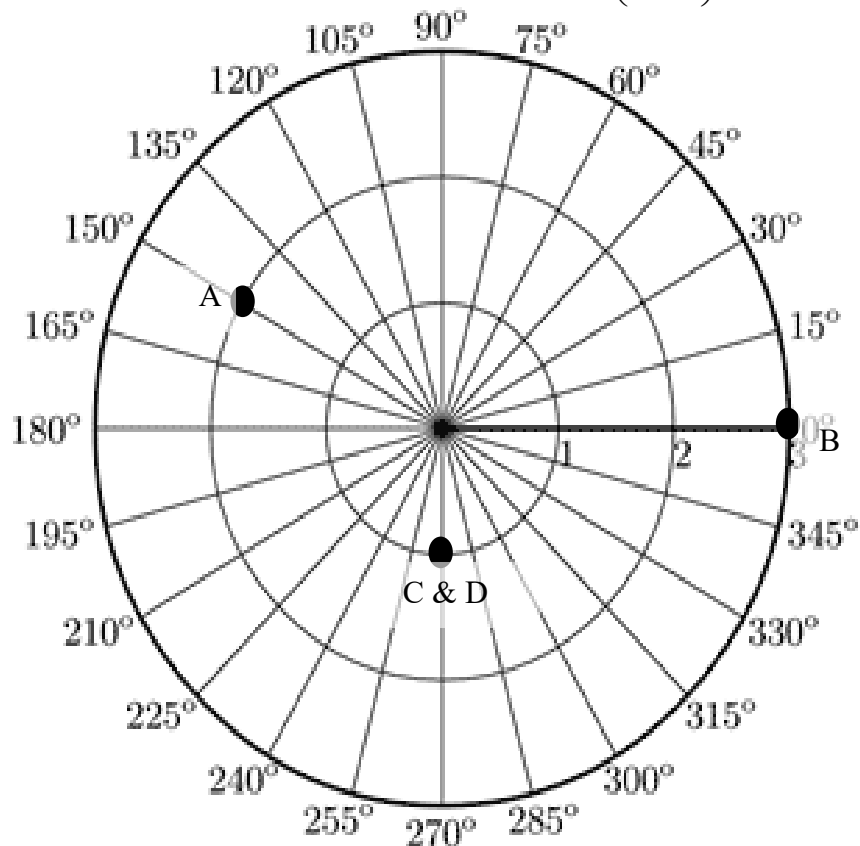
Example 1: Plot the points in the polar coordinate system provided below.

a. $A \left(2, \frac{5\pi}{6} \right)$

b. $B (-3, \pi)$

c. $C \left(1, -\frac{\pi}{2} \right)$

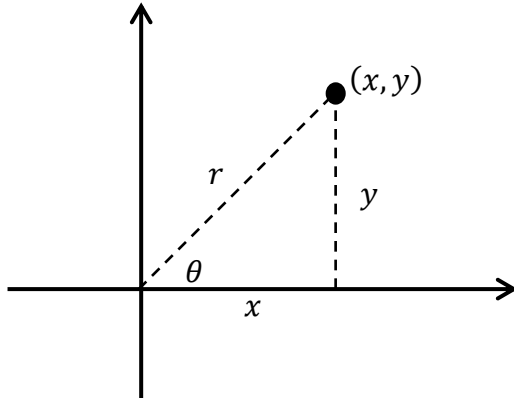
d. $D (-1, 450^\circ)$



Note: There are infinitely many ways to write the coordinates of a point in polar form!!!

To form the same point (r, θ) with different coordinates we simply use an angle that is coterminal with θ .

Polar –Rectangular Conversions



Converting from polar to rectangular

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

Converting from rectangular to polar

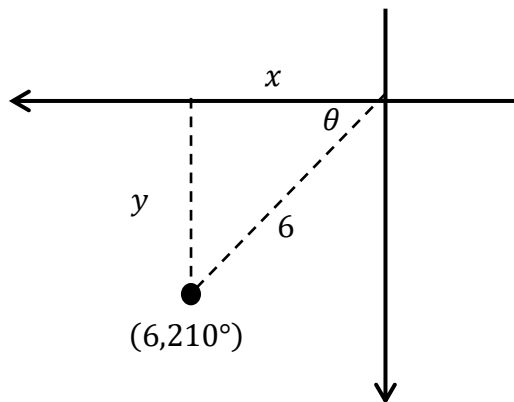
$$r^2 = x^2 + y^2$$

θ is any angle in standard position whose terminal side contains (x, y) .

$$\tan(\theta) = \frac{y}{x} \quad (\theta \text{ is a reference angle})$$

...adjustments may need to be made)

Example 2: Convert $(6, 210^\circ)$ to rectangular coordinates.



$$x = 6 \cos(210^\circ) = 6 \left(-\frac{\sqrt{3}}{2} \right) = -3\sqrt{3}$$

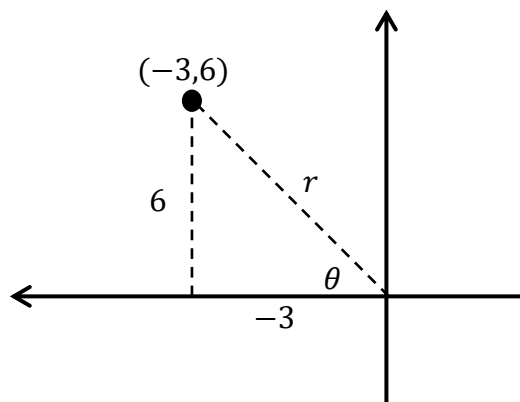
$$y = 6 \sin(210^\circ) = 6 \left(-\frac{1}{2} \right) = -3$$

The rectangular coordinates of the point are:

$$(-3\sqrt{3}, -3)$$

Note that here there is only **1 unique solution**.

Example 3: Convert $(-3, 6)$ to polar coordinates.



$$r = \sqrt{(-3)^2 + (6)^2} = \sqrt{9 + 36} = \sqrt{45} = 3\sqrt{5}$$

$$\sin(\theta) = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}} \quad \text{then we find } \sin^{-1}\left(\frac{2}{\sqrt{5}}\right) \approx 63.4^\circ$$

Notice that this cannot be θ since θ is in quadrant II.
 $\theta \approx 180^\circ - 63.4^\circ = 116.6^\circ$

The polar coordinates of the point are approximately:

$$(3\sqrt{5}, 116.6^\circ)$$

OR

$$(3\sqrt{5}, -243.4^\circ) \quad \text{OR} \quad (-3\sqrt{5}, -63.4^\circ) \quad \text{OR} \quad \dots$$

Note that here there are **infinitely many solutions**.

Converting Equations

Formulas for converting equations:

$$r^2 = x^2 + y^2$$

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

Common Techniques:

- (1) Multiplying both sides of the equation by r .
- (2) Squaring both sides of the equation.

Example 4: Write an equivalent rectangular equation for the polar equation: $r = 2 \cos(\theta)$.

$$r = 2 \cos(\theta) \quad \leftrightarrow \quad r^2 = 2r \cos(\theta) \quad \leftrightarrow \quad x^2 + y^2 = 2x$$

We are done.

However, since we can graph this equation and see it resembles a circle we may wish to continue to manipulate the rectangular equation to make it look like an equation for a circle.

$$x^2 - 2x + y^2 = 0$$

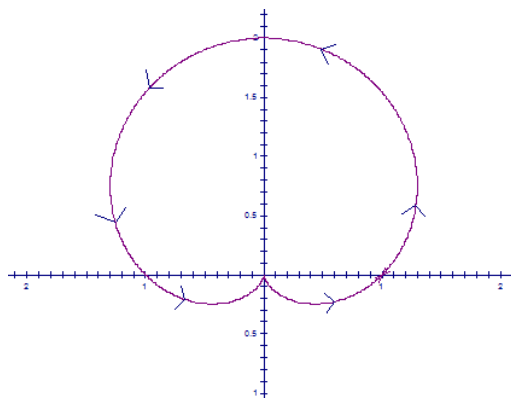
$$x^2 - 2x + 1 + y^2 = 1$$

$$(x - 1)^2 + y^2 = 1$$

We see that this is a circle centered at $(1, 0)$ with radius $r = \sqrt{1} = 1$.

Example 5: Use your calculator to sketch a graph of the polar curve given by $r = 1 + \sin(\theta)$.

A sketch would be:

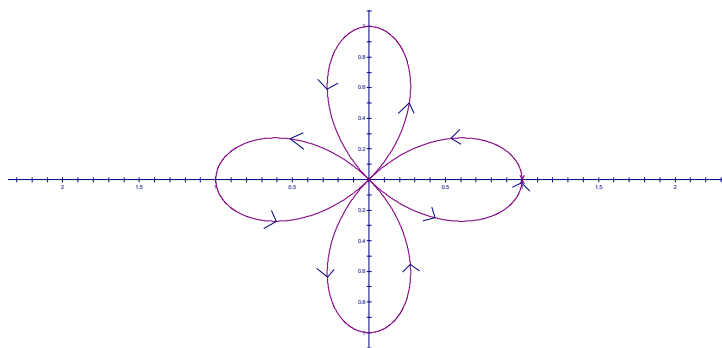


Note: This is called a cardioid and it is completely traced as theta varies from 0 to 2π .

Note: We added arrows to indicate how the curve is traced. This will become important later.

Example 6: Use your calculator to sketch a graph of the polar curve given by $r = \cos(2\theta)$.

A sketch would be:



Note: This is called a 4 leaf rose. The first half of the first leaf is created as theta varies from 0 to $\frac{\pi}{4}$.

Example 7 (you try):

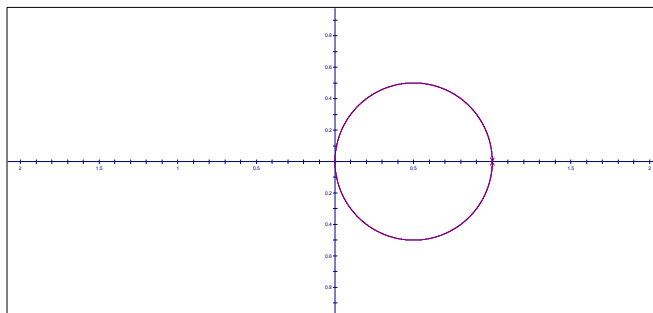
- Convert the point $\left(2, \frac{\pi}{3}\right)$ from polar to rectangular (Cartesian) coordinates.
- Represent the point with rectangular coordinates $(1, -1)$ in terms of polar coordinates.
- Use your calculator to graph $r = \cos(\theta)$ and sketch the graph your calculator produces.
- Convert the equation $r = \cos(\theta)$ from polar to rectangular form.

Solution:

(a) We see the answer is $\left(2 \cos\left(\frac{\pi}{3}\right), 2 \sin\left(\frac{\pi}{3}\right)\right) = (1, \sqrt{3})$.

(b) We note that $r = \sqrt{2}$ and $\tan(\theta) = -1$. Since the point is in the fourth quadrant we take $\theta = \frac{-\pi}{4}$ (there are many possible choices), and a possible answer is $\left(\sqrt{2}, \frac{-\pi}{4}\right)$.

(c) A sketch is:



Note: Interestingly this curve is traced as θ varies from 0 to π .

(d) We note that $r = \cos(\theta) \Leftrightarrow r^2 = r \cos(\theta) \Leftrightarrow x^2 + y^2 = x$.

Note: With some manipulation one can see that the rectangular form equation is the equation of a circle centered at $(0.5, 0)$.

Symmetry in Polar Graphs (Optional Material)

Theorem: (Symmetry Tests)

A polar graph is...

- (a) symmetric about the x -axis if replacing θ by $-\theta$ produces an equivalent equation.
- (b) symmetric about the y -axis if replacing θ by $\pi - \theta$ produces an equivalent equation.
- (c) symmetric about the origin if replacing θ by $\theta + \pi$ (or r by $-r$) produces an equivalent equation.

Example 8: Show that $r = \cos(2\theta)$ is symmetric about the x -axis, the y -axis, and the origin.

Symmetric about the x -axis:

Original equation: $r = \cos(2\theta)$

Replace θ by $-\theta$: $r = \cos(-2\theta) = \cos(2\theta)$

Therefore, $r = \cos(2\theta)$ is symmetric about the x -axis.

Symmetric about the y -axis:

Original equation: $r = \cos(2\theta)$

Replace θ by $\pi - \theta$: $r = \cos[2(\pi - \theta)] = \cos(2\pi - 2\theta) = \cos(2\pi)\cos(2\theta) + \sin(2\pi)\sin(2\theta) = \cos(2\theta)$

Therefore, $r = \cos(2\theta)$ is symmetric about the y -axis.

Symmetric about the origin:

Original equation: $r = \cos(2\theta)$

Replace θ by $\theta + \pi$: $r = \cos[2(\theta + \pi)] = \cos(2\theta + 2\pi) = \cos(2\theta)\cos(2\pi) - \sin(2\theta)\sin(2\pi) = \cos(2\theta)$

Therefore, $r = \cos(2\theta)$ is symmetric about the origin.

